25900262 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBOTA03 - FUNDAMENTALS OF PLANT BIOTECHNOLOGY & INDUSTRIAL BOTANY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or two Sentences

Answer any **10** questions

Each question carries **2** marks

1. Recall any two biosafety concerns of GMOs. [U] / [CO1]
2. What is cryopreservation? [U] / [CO1]
3. Write a short note on plant biotechnology. [U] / [CO1]
4. What is an explant? [U] / [CO2]
5. How does polymeric encapsulation differ from alginate encapsulation in synthetic seed production? [An] / [CO3]
6. Name two causes of somaclonal variation. [U] / [CO3]
7. What is the primary purpose of a plant nursery? [U] / [CO4]
8. How do you control temperature and humidity in a nursery? [E] / [CO4]
9. Define Floriculture. [U] / [CO4]
10. What crops are suitable for organic farming? [U] / [CO5]

11. Describe the nutritional benefits of straw mushrooms. [U] / [CO5]

12. Write short notes on microgreens. [K] / [CO5]

[2x10 = 20]

Part B

Short Answer

Answer any **8** questions

Each question carries **5** marks

13. Genetic engineering is influential in developing novel modifications in plants. Explain. [U] / [CO1]

14. Describe the importance of differentiation, dedifferentiation and redifferentiation in plant tissue culture. [U] / [CO1]

15. Enumerate the applications of Agricultural Biotechnology. [A] / [CO1]

16. Explain the significance of protoplast fusion in the creation of somatic hybrids and cybrids. [U] / [CO3]

17. Analyse the limitations of synthetic seed technology in field-level propagation. [An] / [CO3]

18. What is the importance of callus initiation in micropropagation? [U] / [CO2]

19. Why are seed industries important for agriculture? [A] / [CO5]

20. Dry flowers are good prospects for revenue generation. Validate the statement with examples. [U] / [CO4]

21. Describe the role of seed processing for seed industry. [U] / [CO5]

22. Describe a method for controlling pests and diseases in mushroom production. [A] / [CO5]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Discuss the techniques and significance of somatic hybridization and cybridization. [A] / [CO3]

24. What is organic farming? Discuss the benefits and challenges of organic farming. [An] / [CO5]

[10x1 = 10]